

AutoAIM Studio

Manual de Usuario

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Tested With: 1024+ automated tests (0 failures, 75 skipped)

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Introduccion

AutoAIM Studio is a comprehensive machine learning platform designed for materials science applications. It provides an intuitive graphical interface for data preprocessing, model training, hyperparameter optimization, neural network construction, ensemble modeling, and model interpretation.

Caracteristicas Principales

- Vincule Crystal Structures (v2.0): Conecte CIF structure files to tabular datasets via an ID column
- Data Health Marque (v2.0): Automated quality analysis before training
- Exporteacion de Sesioner (v2.0): Exporte complete results in .txt/.md/.csv/.json
- Intervalos de Confianza (v2.0): Predigaion intervals for regression models
- Tarjetas de Modelo (v2.0): Auto-generated documentation for each saved model
- Automated Data Preprocessing: Handle missing values, categorical encoding, and feature scaling

- Custom Composition Feature Engineering: Generate 106 composition-based descriptors from chemical formulas
- Crystal Structure Loading: Importe CIF, POSCAR, CONTCAR, VASP, XYZ files (v1.2+)
- Multiple ML Algorithms: Bosque Aleatorio, Gradient Boosting, Histogram Gradient Boosting, XGBoost, LightGBM, CatBoost, SVR, Ridge, k-Nearest Neighbors, Proceso Gaussiano (v2.1 — full kernel family tuning + native uncertainty quantification)
- Neural Network Compileer: Visual architecture design with customizable layers
- Hyperparameter Optimization: Bayesian optimization with Optuna, with parameter selection
- Ensemble Modeling: Combine multiple models with weight optimization
- Model Interpretability: SHAP values, permutation importance, partial dependence plots
- Standalone Predigaion Mode: Deploy trained models for inference on new data

Requisitos del Sistema

- Ventanas 10/11 (64-bit)
- 8GB RAM minimum (16GB recommended)
- CUDA-compatible GPU (optional, for neural network acceleration)

Configuraciones Probadas

AutoAIM Studio v2.0.0 has been validated with:

- 1024+ automated tests — 0 failures, 75 skipped (optional dependencies)
- 44 test modules covering core functionality, integration workflows, and utilities
- Python 3.10–3.12 on Ventanas 10/11 and Linux (Ubuntu 22.04)
- PyInstaleer 6.2.0 one-directory builds

Novedades en v2.0.0

Vincule Crystal Structures

The most significant addition in v2.0.0 is the ability to link crystal structure files (CIF) to tabular datasets via an ID column. This bridges the gap between tabular machine learning and structural information.

Como usar:

16. Prepare your CSV with an ID column that matches your CIF filenames (e.g., material_id)
17. Name CIF files as {ID_value}.cif (e.g., mp-1234.cif)
18. Haga clic "Vincule Crystal Structures" in the Data tab
19. Seleccione your ID column and CIF folder
20. AutoAIM extracts 149 crystal descriptors and merges them with your dataset

Crystal features extracted (149 total):

- Lattice (10): cell parameters, volume, symmetry flags (cubic/hexagonal)
- Composition-weighted (13): averaged periodic properties (electronegativity, ionization energy, atomic radius, etc.)
- Structural (3): volume per atom, packing fraction, space group number
- Derived (2): density, total electrons
- Magpie composition (106): full statistical descriptors from the CIF formula
- Advanced structural (15): bond lengths, coordination numbers, RDF peaks, structure complexity index, lattice anisotropy

Todos features use the same computation core as direct CIF loading — only the column prefix differs (crystal_ for linked features).

Polymorph detection: Si multiple rows share the same ID, AutoAIM warns you about potential polymorphs before proceeding.

Importeante: Cuando crystal structures are linked, the Magpie composition featurizer is automatically disabled to avoid duplicate columns. The CIF-derived features already include composition descriptors.

Data Health Marque

Antes de entrenar, run the Data Health Marque to analyze your dataset quality:

- Detects missing values, outliers, duplicate rows, constant columns
- Marques scaling mismatches, high correlations, class imbalance
- Provides an overall score (0-100) and actionable recommendations
- Access via the "Data Health Marque" button in the Data tab

Exporteacion de Sesioner

Exporte your complete session with all results:

- 4 formats: .txt (readable), .md (Markdown), .csv (spreadsheet), .json (machine)
- Includes: model parameters, per-fold CV scores, optimization trials, NN history
- Includes learning curves data, confidence intervals, and optimization configuration
- Access via the "Exporte Session" button in the Resultados tab

Intervalos de Confianza

For regression predictions, AutoAIM can compute confidence intervals:

- 68% and 95% intervals based on cross-validation residuals
- Available in the Prediga tab via the "Include Intervalos de Confianza" checkbox
- Works for all model types (sklearn, neural networks, ensembles)

Tarjetas de Modelo

Each saved model now includes an auto-generated MODEL_CARD.md with:

- Model details, intended use, dataset characteristics
- Performance metrics, confidence intervals, feature names
- Limitations and reproducibility instructions

Pipeline de Fuga de Datos Cero (v2.0 CORRECCION CRITICA)

AutoAIM Studio now uses sklearn Pipeline internally to completely eliminate data leakage during cross-validation:

The problem (now fixed): Anteriorly, StandardScaler was fitted on the entire training set before cross-validation. This meant every validation fold's statistics leaked into the scaler, producing slightly over-optimistic CV scores.

The fix:

- Data is kept raw (unscaled) through the entire workflow
- A Pipeline([preprocessor, model]) is built automatically when you train
- cross_validate() refits the preprocessor inside each fold using only the training data of that fold
- Your validation fold never influences the scaler statistics

Que this means for you:

- CV scores are now scientifically valid — no hidden leakage
- Saved models are complete Pipelines that accept raw DataFrames for inference
- Todos preprocessing happens automatically and correctly behind the scenes
- The 20% internal test split and external validation metrics were already correct and are unchanged

Validacion Externa

Cuando you load an external validation dataset:

- Models report metrics on BOTH internal test split AND external validation set
- Three new columns in Resultados: "Val. RMSE/Acc", "Val. MAE", "Val. R2/F1"
- No data leakage — external data is NEVER used for training

Primeros Pasos

Iniciar la Aplicacion

21. Doble-click the AutoAIM Studio.exe file
22. The main window will open with multiple tabs at the top
23. Inicie by loading your data in the Data tab

Resumen del Flujo de Trabajo

The typical workflow follows these steps:

24. Cargue Data → Pestana de Datos
25. Vincule Crystal Structures → Pestana de Datos (optional, v2.0 — connect CIF files)
26. Health Marque → Pestana de Datos (optional, v2.0 — analyze data quality)
27. Preprocess → Pestana de Datos (handle missing values, encode categoricals)
28. Generate Features → Pestana de Datos (optional, composition descriptors or crystal features)
29. Entrene Models → Pestana de Entrenamiento or Pestana de Redes Neuronales
30. Optimice → Pestana de Optimizacion (optional, for hyperparameter tuning)
31. Cree Ensembles → Pestana de Ensamblado (optional, combine models)
32. Analice Resultados → Pestana de Resultados
33. Exporte Session → Pestana de Resultados (v2.0 — save complete report)
34. Interpret Models → Pestana de Explicabilidad
35. Make Predigaions → Pestana de Prediccion (v2.0 — with confidence intervals)

Pestana de Datos

The Pestana de Datos is your starting point for loading and preparing your dataset.

Carga de Datos

36. Haga clic "Cargue Data" button
37. Seleccione your data file (CSV or Excel format supported)
38. The data preview will appear in the table below

Supported Formats:

- CSV files (.csv)
- Excel files (.xlsx, .xls)

Vista Previa de Datos

Despues loading, you'll see:

- Shape: Number of rows and columns
- Columns: Lista of all column names
- Data Types: Automatic detection of numerical and categorical columns
- Sample Data: Primero few rows of your dataset

Seleccion de Columna Objetivo

39. Seleccione your target variable from the "Target Column" dropdown
40. The system will automatically detect if it's a regression or classification problem

Vincular Estructuras Cristalinas (v2.0)

Conecte crystal structure files (CIF) to your tabular dataset.

Prerequisites:

- Your CSV must have an ID column that matches your CIF filenames
- CIF files must be named {ID_value}.cif (e.g., if ID is mp-1234, the file must be mp-1234.cif)

Pasos:

41. Haga clic "Vincule Crystal Structures" button
42. AutoAIM shows ID column candidates (columns with mostly unique values)
43. Seleccione your ID column from the list
44. Elija the folder containing your CIF files
45. AutoAIM reports: matched CIFs, missing CIFs, and duplicate IDs
46. Haga clic Yes to proceed

Despues linking:

- 45 crystal features are added (11 structural + 34 periodic compositional including oxidation states)
- The Magpie "Generate composition features" checkbox is disabled (composition already included)
- The ID column is automatically excluded from training features

Vincular Estructuras Moleculares (v2.0)

Conecte molecular structure files (.xyz) to your tabular dataset.

Que is a molecule vs. a crystal?

- Molecules are individual, isolated structures (e.g., H_2O, caffeine, a protein ligand) — they have no periodic boundary, no unit cell
- Crystals are periodic structures with a unit cell that repeats in 3D space (e.g., SiO_2 quartz, Fe_2O_3 hematite)
- Si your .xyz file has a unit cell, use Vincule Crystal Structures instead

Prerequisites:

- Your CSV must have an ID column that matches your .xyz filenames (without extension)
- .xyz files must be named {ID_value}.xyz

Pasos:

47. Haga clic "Vincule Molecular Structures" button
48. Seleccione your ID column (the one matching .xyz filenames)
49. Elija the folder containing your .xyz files
50. AutoAIM reports: matched files, missing files
51. Haga clic Yes to proceed

Despues linking:

- 149 molecular features are added (composition + geometry 3D + topology + 34 periodic properties + oxidation states)
- The ID column is automatically excluded from training features

Alternative: Direct training on .xyz files

- Si you only have .xyz files and no CSV, click "Cargue Molecular Structures" or "Cargue Molecular Directory"
- The extracted molecular features become your full dataset
- Seleccione a target column (if available) and train directly

Verificacion de Calidad de Datos (v2.0)

Analice your dataset quality before training.

Pasos:

52. Haga clic "Data Health Marque" button
53. Review the quality score (0-100) and severity level
54. Marque detected issues and recommendations
55. Fix issues if needed before training

Marques performed:

- Missing values (severity based on percentage)
- Duplicate rows
- Constant columns
- Outliers (IQR method)
- Scaling mismatches between features
- Alto correlations between features (>90%)
- Sample size adequacy
- Class imbalance (for classification)
- Target column quality

Opciones de Preprocesamiento

Manejar Valores Faltantes

- Marquebox: "Handle missing values"
- Cuando enabled, missing numerical values are imputed with the median
- Missing categorical values are imputed with the most frequent value

Codificar Variables Categoricas

- Marquebox: "Encode categorical variables"
- Cuando enabled, categorical variables are one-hot encoded
- This expands categorical columns into multiple binary columns

Escalar Caracteristicas

- Marquebox: "Scale features"
- Cuando enabled, features are standardized (zero mean, unit variance)
- Recomendado for algorithms sensitive to feature scales (SVR, Redes Neuronales)

Generar Caracteristicas de Composicion

- Marquebox: "Generate composition features (106 descriptors)"
- Requires: A column containing chemical formulas (e.g., "Fe2O3", "SiO2")
- Cuando enabled, the system extracts 106 composition-based descriptors including:
- Atomic number statistics (mean, std, min, max)
- Atomic mass statistics
- Electronegativity statistics
- Covalent radius statistics
- Ionization energy statistics
- And many more material properties

Nota: The composition feature generator is a custom implementation designed for materials science applications. It analyzes chemical formulas and computes aggregate statistics based on elemental properties.

Preparacion de Datos

Despues configuring options:

56. Haga clic "Prepare Data" button
57. The system will split data into train/test sets (80/20 by default)
58. Todos selected preprocessing steps are applied
59. A confirmation message will appear when complete

Pestana de Entrenamiento

The Pestana de Entrenamiento allows you to train multiple machine learning models simultaneously.

Seleccion de Modelo

Seleccione one or more models from the list:

- Bosque Aleatorio: Ensemble of decision trees, good for general-purpose prediction
- Gradient Boosting: Sequential ensemble, often achieves high accuracy (sequential — slower on large datasets)
- Histogram Gradient Boosting: v2.0 — Parallel gradient boosting via histograms. 10-100x faster than classic GB on large datasets. Recomendado for gradient boosting workloads.
- XGBoost: Optimiced gradient boosting, industry standard

- LightGBM: Fast gradient boosting with leaf-wise growth
- CatBoost: Gradient boosting optimized for categorical features
- SVR: Support Vector Regression, good for smaller datasets
- Ridge: Linear regression with L2 regularization, simple baseline
- Proceso Gaussiano (v2.1): Probabilistic regressor with native uncertainty quantification (±1σ posterior standard deviation) and full kernel family tuning (RBF, Matern, RationalQuadratic, WhiteKernel, RBF+WhiteKernel). Best for small-to-medium datasets where prediction confidence matters. Kernel hyperparameters are fully optimizable via Bayesian optimization.

Configuracion de Entrenamiento

- Problem Escriba: Automatically detected (regression/classification)
- CV Strategy: Como cross-validation splits your data (default: K-Fold). Vea Elegir una Estrategia de VC below. Options: K-Fold, Stratified K-Fold (quantiles), Agrupe K-Fold, Leave-One-Out (LOO), Leave-One-Agrupe-Out (LOGO).
- Cross-Validation Folds: Number of CV folds (default: 5). Deshabilited for LOO/LOGO (folds are set automatically).
- Random State: Seed for reproducibility (default: 42). Controls data split + model init + CV folds. Vea Semilla Aleatoria y Reproducibilidad below.

Proceso de Entrenamiento

60. Seleccione desired models (check the boxes)
61. Haga clic "Entrene Seleccioneed Models"
62. Progress will be shown in the status bar
63. Resultados will appear in the Pestana de Resultados automatically

Entrenamiento Manual con Parametros Personalizados

For advanced users who want full control over model hyperparameters:

64. Haga clic on a model name in the list (not the checkbox)
65. The Manual Parameter Cuadro de dialogo opens with all available parameters
66. Establezca your custom values (e.g., n_estimators, max_depth)
67. Configure common parameters:

- random_state: Solo shown if the model accepts it (hidden for deterministic models like SVR, Ridge)
- n_jobs: Solo shown if the model supports parallel training (hidden for single-threaded models)
- cv_folds: Configurable cross-validation folds (v2.0 — was hardcoded to 5). Deshabilited when LOO/LOGO is selected.
- cv_strategy: Full cross-validation configuration via the shared CV widget (v2.0) — K-Fold, Stratified K-Fold (quantiles), Agrupe K-Fold, LOO, or LOGO, with bins / group source / clusters / group column shown conditionally. Vea Elegir una Estrategia de VC below.

68. Haga clic Entrene

v2.0 Nota: Deterministic models (SVR, Ridge) don't show random_state or n_jobs in the dialog since they don't accept these parameters. This prevents confusion when setting values that have no effect.

v2.0 Nota: Cuando LOO-CV or LOGO is selected (in the Entreneing tab or the Manual Parameter Cuadro de dialogo), the CV Folds spinner is automatically disabled and an info message reminds you that the folds are set automatically (N samples for LOO, N groups for LOGO).

Elegir una Estrategia de VC (v2.0)

Cross-validation estimates how well a model generalizes. The right strategy depends on your dataset size, whether your target is skewed, and whether your samples form groups that could "leak" information between train and test. AutoAIM supports five strategies, available consistently across sklearn regressors, Redes Neuronales, and Ensembles; the same controls appear in the Entreneing tab, the Optimization tab, the Neural Network tab, the Ensemble tab, and the Manual Parameter dialog. (Quantile stratification and the reserved bin are regression-only; classification folds are stratified by class automatically.)

Your situation	Recomendado strategy	Por que
General case, N > 30, roughly balanced target	K-Fold	Standard, fast, robust
Skewed target (e.g. most Tc low, a few high)	Stratified K-Fold (quantiles)	Every fold preserves the target distribution
Samples form chemical families / batches / synthesis groups	Agrupe K-Fold	A group never crosses train/test → no leakage
Muy small dataset (N ≤ 30)	Leave-One-Out (LOO)	Maximizes training data per fold
Pocos groups, want strict group held-out test	Leave-One-Agrupe-Out (LOGO)	Each group tested once; folds = n_groups

The five strategies:

69. K-Fold (default). Random folds. For classification, folds are automatically stratified by class.
70. Stratified K-Fold (quantiles) — regression only. The target is binned into quantiles and each fold preserves the target distribution. Use it for skewed targets. Not available for classification (which is already class-stratified under K-Fold).

Como the Quantile Bins control works: Cuando you select this strategy, a Quantile Bins spinner appears (only for this strategy), with range 2–20 and default 5. The target is split into that many

quantiles (equal-count bins — with 5 bins, each holds ~20% of the samples sorted by target value). Each bin becomes a pseudo-class, and StratifiedKFold then gives every fold a proportional share of every bin, so no fold ends up with only low or only high values.

The bin count is a request, not a guarantee. AutoAIM enforces a hard rule — each bin must contain at least as many samples as there are folds (otherwise a fold could miss a bin entirely). So the effective maximum is N ÷ folds (with N=50 and 5 folds, at most 10 bins). Si your target has many repeated values or the request exceeds what the data supports, the bin count is reduced automatically and an orange info note tells you exactly what was used (e.g. "Quantile bins reduced from 10 to 5 ..."). Si the target is constant, or not even a 2-bin median split is possible, the run is blocked with a message suggesting standard K-Fold. The session report header shows the requested bins and any reduction as a warning line.

Reserve a bin for an exact value (v2.0). Muchos materials targets have a degenerate value meaning "property absent": Tc = 0 K (non-superconductor), Eg = 0 eV (metal), E_hull = 0 (stable phase), activity = 0 (inactive catalyst). These are point masses, not continuous values, and if they land unevenly across folds the per-fold scores become hard to compare. Enabling "Reserve a bin for an exact value" (visible only for this strategy) isolates every sample matching that value into its own stratum, so every fold carries the same proportion of it; the remaining samples are binned into the other bins − 1 quantiles.

- Special Value: the exact target value to isolate (e.g. 0.0).
- Tolerance: how close a value must be to count (default 0.0 = exact match). Increase it only if your data stores the value as "below detection threshold" (e.g. 0.01 instead of 0).
- Auto-detection: if AutoAIM detects a dominant repeated value in your target (at least ~5% of samples and at least 2 per fold), a cyan suggestion appears — e.g. "Detected 25 samples with target = 0 (28%). This usually means 'property absent'. Consider enabling 'Reserve a bin...'". The suggestion is guidance only; nothing is applied until you enable the checkbox.
- Manual use is validated with real numbers, not a generic warning. Si your chosen value matches too few samples to stratify (fewer than the number of folds), the run is blocked. Si it is the majority (>50%) or a very small fraction (<5%), a warning shows the actual count and percentage and asks you to confirm — so you can reconsider whether the reserved bin is meaningful for your data.

Nota: a reserved bin improves the honesty of the CV estimate; it does not change the underlying modeling. For a heavily zero-inflated target, the more rigorous approach is a two-stage model (classify "does it exhibit the property?" then regress "by how much"), which is outside the current scope.

71. Agrupe K-Fold. You define groups; samples from the same group never appear on both sides of a train/test split. Elija the grouping source:

- Chemical system — same set of elements (all Y-Ba-Cu-O together). The standard anti-leakage split for materials data.

- Exact formula — same stoichiometry together (reduced form, so $Y_2Ba_4Cu_6O_{14}$ groups with $YBa_2Cu_3O_7$).
- Clustering (advanced) — agglomerative clustering over element fractions (deterministic; set the cluster count).
- Dataset column — any column: synthesis batch, laboratory, source, year.

72. Leave-One-Out (LOO). Each sample left out once; folds = N. Blocked for N > 100; warns for 30 < N ≤ 100.
73. Leave-One-Agrupe-Out (LOGO). Each group left out once; folds = number of groups. Needs ≥ 3 groups; blocked for > 100 groups; warns for 30 < n_groups ≤ 100.

Metrics note: For LOO and LOGO (and any grouped fold that holds a single sample), R^2 is mathematically undefined for a 1-sample test fold. AutoAIM computes a global R^2 over all out-of-fold predictions and reports per-sample / per-group MAE. For K-Fold, Stratified, and Agrupe K-Fold it reports per-fold Entrene Score + Pruebe Score (plus a Pruebe Size column when folds are uneven, as in Agrupe K-Fold).

Advertencias and blocks: AutoAIM validates your choice before running and will block impossible combinations (e.g. more folds than groups, LOGO with < 3 groups, a group column with missing values) with a clear message, and warn (asking for confirmation) about expensive or risky ones (LOO/LOGO cost, highly imbalanced groups).

Comprender los Resultados

Despues training, each model shows:

- RMSE: Root Mean Squared Error (lower is better)
- MAE: Mean Absolute Error (lower is better)
- R^2: Coefficient of determination (higher is better, max 1.0)
- Entreneing Time: Como long the training took

Ventana Emergente de Detalles — Puntuaciones de Validacion Cruzada

Haga clic "Details" on any trained model to see the full cross-validation breakdown:

K-Fold CV (default):

```
Fold |  Entrene Score |   Pruebe Score
-----+--------------+-------------
   1 |     0.999960 |     0.975628
   2 |     0.999960 |     0.973907
 ...
Mean |     0.999962 |     0.949832
 Std |     0.000003 |     0.050600
```

Agrupe K-Fold (adds a Pruebe Size column because grouped folds are uneven):

```
Fold |  Entrene Score |   Pruebe Score | Pruebe Size
-----+--------------+--------------+----------
   1 |     0.999960 |     0.975628 |        12
   2 |     0.999960 |     0.973907 |         8
 ...
Mean |     0.999962 |     0.949832 |
 Std |     0.000003 |     0.050600 |
```

Leave-One-Out (LOO-CV):

```
R² (global): 0.960446

Per-sample Mean Absolute Error (MAE):
Sample  1: 0.019916
Sample  2: 0.003685
 ...
Mean MAE ± Std: 0.012872 ± 0.008515
```

Leave-One-Agrupe-Out (LOGO):

```
R² (global): 0.978000

Per-group Mean Absolute Error (MAE):
Agrupe  1: 0.120000
Agrupe  2: 0.110000
 ...
```

Importeante: For LOO-CV and LOGO, R^2 per fold is mathematically undefined (each fold has a single test sample or group). AutoAIM computes a global R^2 over all out-of-fold predictions and reports per-sample / per-group MAE. This is the correct scientific approach for these strategies.

Semilla Aleatoria y Reproducibilidad (v2.0)

AutoAIM lets you control the random seed everywhere randomness appears, so results are fully reproducible. Cuando you set a Random State / Random Seed, AutoAIM applies it to all of the following at once:

- Entrene/test split (Entreneing tab "Random State")
- Model weight initialization / bootstrap sampling (stochastic models)
- CV fold shuffling (the fold assignment follows your seed)
- Optimicer search sequence (Bayesian sampler seed; Cuadricula Busque is deterministic)

Donde configurarla

Tab	Control	Que it controls

Entreneing	Random State	data split + model init + CV folds
Optimization	Random Seed	model init + search sampler + CV folds
Neural Network	Random Seed (training) / Random Seed (optimization)	weight init + data shuffling + CV folds
Ensemble	Random Seed (training) / Random Seed (weight opt)	ensemble CV folds + weight search
Manual Parameter dialog	random_state	model init + CV folds (shown only for models that accept it)

Que modelos la utilizan

Stochastic models honor the seed: Bosque Aleatorio, Gradient Boosting, Histogram Gradient Boosting, XGBoost, LightGBM, CatBoost, Redes Neuronales, Ensembles. Deterministic models (SVR, Ridge, k-NN) do not — they have no randomness to seed, so the control is disabled/hidden for them and setting a value has no effect on the model itself (it still affects the CV split shuffling).

Notas

- CatBoost uses random_seed internally; AutoAIM maps your Random State to it automatically (this was a silent bug fixed in v2.0 — earlier the optimizer passed random_state, which CatBoost ignores).
- Same seed + same data + same settings → identical results, guaranteed. This is covered by automated reproducibility tests (test_random_state.py).

Proceso Gaussiano (v2.0 → v2.1)

Proceso Gaussiano (GP) is a probabilistic machine learning model that provides predictions together with a native uncertainty estimate — the posterior standard deviation. This makes it uniquely valuable for materials science, where knowing "how confident is the model?" is often as important as the prediction itself.

v2.1: Ajuste de Familia de Kernel

AutoAIM Studio v2.0.0 introduces full kernel family support — the most scientifically complete GP implementation in the AutoML ecosystem for materials science.

Familias de Kernel Soportadas

Kernel	Description	Best For
RBF	Infinitely differentiable, smooth functions	General-purpose, well-behaved data
Matern	Controlled smoothness via nu (0.5, 1.5, 2.5)	Data with known roughness; 0.5 = once differentiable (exponential), 2.5 = smooth (almost RBF)
RationalQuadratic	Mixture of RBFs with different length-scales	Multi-scale data, varying correlation ranges
WhiteKernel	Independent observation noise	Explicit noise modeling, heteroscedastic data
RBF + WhiteKernel	Signal + noise decomposition	Separating signal covariance from noise covariance

Hiperparametros Ajustables

Parameter	Escriba	Range	Description
kernel_type	categorical	RBF, Matern, RQ, White, RBF+White	Kernel family selector
kernel_length_scale	float_log	1e-3 → 100.0	Characteristic length scale (most important GP hyperparameter)
kernel_nu	categorical	0.5, 1.5, 2.5	Matern smoothness (lower = rougher)
kernel_alpha_rq	float_log	1e-3 → 100.0	Rational Quadratic tail heaviness
kernel_noise_level	float_log	1e-5 → 1.0	WhiteKernel noise magnitude
alpha	float_log	1e-10 → 1.0	Tikhonov regularization (observation noise)
n_restarts_optimizer	int	1 → 20	L-BFGS-B restarts for marginal likelihood

normalize_y	categorical	True, False	Normalize target before fitting

Bayesian optimization (Optuna TPE) is recommended for kernel tuning — it can optimize all 8 parameters simultaneously. Cuadricula Busque supports scalar GP params (alpha, n_restarts_optimizer, normalize_y) but automatically excludes kernel parameters with a warning.

Por que usar Proceso Gaussiano?

Situation	Por que GP excels
Small-to-medium datasets (N < 2,000)	GP interpolates well; uncertainty is largest far from training data
You need prediction confidence	Every prediction comes with ±1σ (standard deviation of the posterior)
Exploring a new chemical space	Alto uncertainty flags regions where the model has little information
Benchmarking / baseline	Simple, theoretically grounded alternative to tree ensembles

Advertencia de Escalabilidad

GP training scales as $O(n^3)$ due to matrix inversion. For datasets above ~2,000 samples, training becomes slow. AutoAIM shows a console warning when GP is selected with N > 2,000. For very large datasets, use a tree-based model instead.

Lo que obtiene

- Predigaion: The mean of the posterior distribution (same as any regressor)
- prediction_std: The standard deviation of the posterior (unique to GP). Lower = more confident.
- Kernel: Configurable kernel family (RBF, Matern, RQ, White, RBF+White) with automatic hyperparameter optimization
- Alpha: Learned noise level (how much the data deviates from a perfect smooth function)

Donde aparece

- Pestana de Entrenamiento: Seleccione "Proceso Gaussiano" from the model list and train normally. Use the "Custom Parametros" button to select a kernel family and tune kernel hyperparameters manually.

- Pestana de Optimizacion: Optimice all GP parameters including kernel_type, kernel_length_scale, kernel_nu, kernel_alpha_rq, kernel_noise_level, alpha, n_restarts_optimizer, and normalize_y. Bayesian optimization is recommended for kernel tuning.
- Pestana de Prediccion: Predigaions include an extra prediction_std column with ±1σ uncertainty
- Model Card: Shows kernel family, fitted kernel string, length scale, nu (Matern), alpha (RQ), noise level, and log marginal likelihood
- Exporteacion de Sesion: GP Metadata section includes full kernel provenance for scientific reproducibility

Interpretacion de prediction_std

prediction_std	Interpretation
< 0.1	Alto confidence — prediction is near training data
0.1 – 0.5	Moderate confidence — some extrapolation
> 0.5	Bajo confidence — far from training data or high noise

Importeante: prediction_std is the model's own epistemic uncertainty, not a confidence interval. For a 95% confidence interval, multiply by ~1.96 (assuming approximately Gaussian posterior).

Parametros

Parameter	Predeterminado	Description
alpha	1e-10	Noise level added to diagonal of kernel matrix
n_restarts_optimizer	10	Number of random restarts for L-BFGS-B kernel optimization
normalize_y	True	Normalize target to zero mean, unit variance before fitting
random_state	42	Seed for optimizer restarts (affects initial points)
kernel_type	RBF	Kernel family: RBF, Matern, RationalQuadratic, WhiteKernel,

		RBF+WhiteKernel
kernel_length_scale	1.0	Characteristic length scale of the kernel (smaller = wigglier, larger = smoother)
kernel_nu	2.5	Matern smoothness: 0.5 (rough), 1.5, 2.5 (smooth, almost RBF)
kernel_alpha_rq	1.0	Rational Quadratic tail heaviness (low = heavy tails, high ≈ RBF)
kernel_noise_level	1e-5	WhiteKernel noise magnitude (explicit observation noise)

Procedencia del Modelo

v2.1: For scientific reproducibility, the Pestana de Resultados Details popup and Exporteacion de Sesion capture the full kernel provenance:

- kernel_family: RBF, Matern, RationalQuadratic, WhiteKernel, or Composite
- kernel_length_scale: Fitted length scale (4 decimal precision)
- kernel_nu: Fitted Matern smoothness (if applicable)
- kernel_alpha: Fitted RQ tail parameter (if applicable)
- kernel_noise_level: Fitted WhiteKernel noise (if applicable)
- log_marginal_likelihood_value: Optimiced marginal log-likelihood (model evidence)

This metadata is essential for publication-ready model cards and ensures that any GP model can be fully reconstructed from the session export.

Soporte SHAP

GP uses KernelExplainer (fallback) because it is not a tree, linear, or neural model. This is slower than TreeExplainer. For datasets > 500 samples, SHAP values may take noticeable time.

Pestana de Redes Neuronales

The Pestana de Redes Neuronales provides a visual interface for designing and training custom neural networks.

v2.0 Layout: The tab is split into two panels. The left panel (architecture configuration, ~75% of screen width) contains all controls: layer settings, training configuration, a Cross-Validation section, hyperparameter optimization, and action buttons (Auto-Optimice, Aplique Best, Entrene NN). The left panel is vertically scrollable, so every control and button stays reachable on typical

laptop and desktop screens — nothing is ever cut off. The right panel (~25% of screen width) shows Entreneing Progress, Vista Previa de Arquitectura, Entreneing Resultados, and Model Information consoles.

Diseno de Arquitectura

Numero de Capas Ocultas

- Use the spinbox to set 1-5 hidden layers
- Each layer can be configured independently

Configuracion de Capas

For each hidden layer, you can set:

- Units: Number of neurons (16-2048)
- Activation Function: ReLU, Leaky ReLU, Tanh, Sigmoid, GELU, ELU
- Dropout Rate: Regularization (0.0-0.5)
- Batch Normalization: Habilite/disable batch normalization

Vista Previa de Arquitectura

Haga clic "Actualice Vista Previa de Arquitectura" to see:

- Input layer dimensions
- Hidden layer configurations
- Output layer dimensions
- Total number of parameters
- Entreneable parameters

Configuracion de Entrenamiento

- Epochs: Number of training iterations (10-10,000)
- Batch Size: Samples per gradient update (8-512)
- Learning Rate: Paso size for optimization (0.00001-0.1)
- Optimicer: Adam, AdamW, SGD, RMSprop
- Weight Decay (L2): Regularization strength (0.0-0.1)
- Temprano Stopping Patience: Detenga if no improvement after N epochs (5-500)

Validacion Cruzada (v2.0)

Redes Neuronales now support the same five CV strategies as sklearn regressors (see Elegir una Estrategia de VC in the Pestana de Entrenamiento): K-Fold, Stratified K-Fold (quantiles, with optional reserved bin), Agrupe K-Fold, LOO, and LOGO. Each CV fold re-trains a fresh copy of the network, so the estimate reflects true generalization. Nota that LOO/LOGO train one network per sample/group — expensive for NNs, so use them only on small datasets; the usual blocks/warnings apply.

Proceso de Entrenamiento

74. Configure architecture and training parameters
75. Haga clic "Entrene Neural Network"
76. Watch training progress:

- Progress bar shows current epoch
- Registre shows train/validation loss per epoch

77. Entreneing stops automatically when:

- Maximum epochs reached, OR
- Temprano stopping triggered (no improvement)

Optimizacion de Arquitectura de RN

v2.0: You can optimize your neural network architecture using Bayesian optimization:

78. Configure initial architecture (this will be the starting point)
79. Haga clic "Optimice Architecture"
80. Establezca optimization parameters:

- Number of Trials: Como many architectures to try
- Max Layers: Maximum number of hidden layers
- Min/Max Units: Range for neurons per layer
- CV Folds: Cross-validation folds for evaluation

81. Haga clic "Inicie Optimization"
82. Despues optimization completes, the best architecture is automatically applied
83. Haga clic "Entrene with Best" to train with the optimized architecture

v2.0 Nota: Optimiced NNs show [OPTIMIZATION DETAILS] in the session export, including the number of trials, best score, and optimization configuration — same as sklearn optimized models.

Guardar Redes Neuronales

Despues training:

84. Haga clic "Guarde Model"
85. Elija format:

- PyTorch Model (.pt): Simple format
- Model Bundle: Includes manifest.json, model, and preprocessor (recommended for Prediga tab)

Pestana de Optimizacion

The Pestana de Optimizacion finds the best hyperparameters for your models using either Bayesian optimization (default) or Cuadricula Busque (exhaustive evaluation).

Seleccion de Estrategia de Busqueda

Bayesian (Optuna) — Recomendado for most cases:

- Smart exploration: tests the most promising combinations first
- Needs fewer trials (50–100) for excellent results
- Supports pruning (early stopping of bad trials)
- Handles continuous and discrete parameters natively
- Best for 3+ parameters or wide ranges

Cuadricula Busque — Deterministic and reproducible:

- Evaluates every combination in the parameter grid
- Same inputs always produce the same result
- Best for ≤3 parameters with small discrete ranges
- Useful as a baseline comparison against Bayesian results
- Elija resolution: Coarse (3 points), Medio (5), or Fine (7) per parameter

AutoAIM shows an estimated combination count before starting Cuadricula Busque. Si the estimate exceeds 5,000, Cuadricula Busque is blocked with a warning.

v2.0 UX: Cuando Bayesian optimization is selected, the Cuadricula Resolution selector is automatically disabled and an orange info label reminds you that "Cuadricula Resolution is only used with Cuadricula Busque. Bayesian optimization uses adaptive sampling, not fixed grids."

Estrategia de Validacion Cruzada

The Optimization tab offers the same five CV strategies as the Entreneing tab (see Elegir una Estrategia de VC in the Pestana de Entrenamiento section): K-Fold, Stratified K-Fold (quantiles), Agrupe K-Fold, LOO, and LOGO. The selected strategy applies to every trial of both Bayesian and Cuadricula Busque optimization — each candidate hyperparameter set is scored with that CV. Agrupeed strategies (Agrupe K-Fold / LOGO) keep group membership intact inside the optimizer, so hyperparameters are selected without chemical/batch leakage.

For LOO and LOGO, optimization uses Mean Absolute Error as the objective (R^2 is undefined for single-sample folds); for the other strategies it uses R^2.

The same blocks and warnings apply (e.g. LOO blocked for N > 100, more folds than groups blocked, LOGO blocked for > 100 groups).

Seleccion de Modelo

Seleccion de Modelo

Seleccione a single model to optimize from the dropdown:

- Bosque Aleatorio
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost
- SVR
- Proceso Gaussiano (v2.1 — full kernel tuning)

Configuracion de Optimizacion

- Busque Strategy: Bayesian (Optuna) or Cuadricula Busque
- Cuadricula Resolution (Cuadricula Busque only): Coarse (3 pts) / Medio (5) / Fine (7) per parameter. Deshabilited when Bayesian is selected.
- Number of Trials (Bayesian only): Como many hyperparameter combinations to try (10–200)
- CV Strategy: K-Fold, Stratified K-Fold (quantiles), Agrupe K-Fold, LOO, or LOGO (see Elegir una Estrategia de VC)
- CV Folds: Cross-validation folds for evaluating each combination. Deshabilited for LOO/LOGO (folds set automatically).
- Agrupe By / Bins / Clusters / Agrupe Column: shown only for grouped or stratified strategies
- Random Seed: For reproducible optimization (model init + search sampler + CV folds). Deshabilited for deterministic models (SVR, k-NN), which accept no seed.
- Timeout: Maximum optimization time in seconds (0 = no limit)
- Habilite Pruning (Bayesian only): Detenga unpromising trials early (recommended)

Rangos Personalizados de Parametros

Haga clic "Edite Parameter Ranges" to customize the search space:

For each parameter, you can set:

- Min Value: Lower bound
- Max Value: Upper bound
- Escriba: Integer or Float

Available Parametros by Model:

Model	Parametros
Bosque Aleatorio	n_estimators, max_depth, min_samples_split, min_samples_leaf
Gradient Boosting	n_estimators, max_depth, learning_rate,

	subsample
Histogram Gradient Boosting	max_iter, max_depth, learning_rate, min_samples_leaf, max_bins, l2_regularization
XGBoost	n_estimators, max_depth, learning_rate, subsample, colsample_bytree, reg_alpha, reg_lambda
LightGBM	n_estimators, max_depth, learning_rate, num_leaves, subsample, colsample_bytree, reg_alpha, reg_lambda
CatBoost	iterations, depth, learning_rate, l2_leaf_reg
SVR	C, epsilon, gamma
Proceso Gaussiano	kernel_type, kernel_length_scale, kernel_nu, kernel_alpha_rq, kernel_noise_level, alpha, n_restarts_optimizer, normalize_y (v2.1)

Haga clic "Reinicie to Predeterminado" to restore original ranges.

Proceso de Optimizacion

86. Seleccione model and configure settings
87. (Opcional) Edite parameter ranges
88. Haga clic "Inicie Optimization"
89. Monitoree progress:

- Progress bar shows current trial
- Registre shows trial results in real-time

Resultados

Despues optimization:

- Best Parametros: Optimal hyperparameter values found
- Best Score: Best performance achieved
- Number of Trials: Total trials completed
- Optimization Time: Total time elapsed

v2.0: Haga clic "Details" to see the per-trial breakdown, or "Entrene with Best" to train a final model. The success message and Details panel now show Entrene Score + Pruebe Score side by side for K-Fold, and R^2 (global) + per-sample MAE for LOO-CV.

Entrenar con Mejores Parametros

90. Despues optimization completes, click "Entrene with Best"
91. The model is trained using the optimal hyperparameters found
92. The trained model appears in the Pestana de Resultados

v2.0 Nota: Multiple optimizations of the same algorithm are now preserved. Si you optimize XGBoost with 5 CV folds and then again with 3 folds, both optimizations remain available. Each trained model links to its correct optimization configuration in the session export.

- Trial History Tabla: Complete history of all trials

Guardar Modelos Optimizados

Haga clic "Guarde Optimiced Model":

93. Elija format:

- Model Bundle: Includes manifest.json (recommended)
- Legacy Format: Joblib/pickle only

94. Seleccione destination folder
95. Model is saved and appears in Pestana de Resultados

Pestana de Resultados

The Pestana de Resultados displays all trained, optimized, and ensemble models.

Lista de Modelos

Todos models are shown in a table with:

- Model Name: Identifier
- Escriba: Entreneed, Optimiced, Neural Network, or Ensemble
- RMSE: Root Mean Squared Error
- MAE: Mean Absolute Error
- R^2: Coefficient of determination

Acciones

For each model, you can:

- Details: Vea detailed information about the model
- Elimine: Remueva the model from the list
- Exporte: Guarde as model bundle or legacy format

Detalles del Modelo

Haga clicing "Details" shows:

- Model type and source
- Entreneing/optimization time
- Todos performance metrics
- Best parameters (for optimized models)
- Feature importance (if available)
- Neural network architecture (for NN models)
- Cross-Validation breakdown (v2.0):
- K-Fold: Entrene Score + Pruebe Score per fold, with separate Mean and Std rows
- LOO-CV: R^2 (global) + per-sample MAE table, with Mean MAE ± Std
- v2.1: Proceso Gaussiano kernel provenance (for GP models):
- Kernel family (RBF, Matern, RationalQuadratic, WhiteKernel, Composite)
- Fitted kernel string, length scale, nu, alpha, noise level
- Registre marginal likelihood (model evidence)
- Predigaion uncertainty statistics (mean, std, min, max of prediction_std)

Curvas de Aprendizaje

Seleccione a model and click "Generate Curvas de Aprendizaje" to visualize:

- Entreneing vs validation loss over epochs
- Helps diagnose overfitting/underfitting

Exportear Modelos

96. Seleccione model from dropdown
97. Haga clic "Exporte Model Bundle"
98. Elija destination folder
99. Bundle contains:

- manifest.json: Model metadata
- pipeline.joblib or model.pt: Entreneed model
- preprocessor.joblib: Preprocessing pipeline (for NN)

Exporteacion de Sesion (v2.0 → v2.1)

Exporte your complete session with all results:

100. Haga clic "Exporte Session" button
101. Elija format:

- .txt: Human-readable report (recommended)
- .md: Markdown for documentation
- .csv: Spreadsheet format
- .json: Machine-readable

102. The export includes:

- Todos trained models with metrics and per-fold CV scores
- Todos optimized models with [OPTIMIZATION DETAILS] (trials, config, best params)
- Todos neural networks with architecture, training history, and CV fold breakdown
- Todos optimized NNs with optimization time and details
- Ensemble results
- Final ranking table
- Reproducibility checklist (random seeds, configs)
- v2.1: Clean model parameters (no Pipeline noise, no model__ prefix)
- v2.1: GP Metadata section with kernel family, fitted hyperparameters, and prediction uncertainty statistics

v2.1 Cleanup: The export now filters out internal Pipeline noise (preprocessor__*, steps, memory, verbose) and strips the model__ prefix from parameter keys. This makes the report significantly cleaner and more readable for both beginners and advanced users.

v2.0 Reliability: The export uses per-result error handling — if one model fails to export, the others still appear. Anteriorly, a single bad result could destroy the entire export.

v2.0 Accuracy: Optimization configurations are read from the stored result (captured at optimization time), not from the current GUI values. This means if you optimized XGBoost with 5 folds, changed to 3 folds, and optimized again, both exports show their correct configurations.

v2.0 NN Optimization Details: Optimiced neural networks (e.g., NN_Optimiced) now show [OPTIMIZATION DETAILS] inline, including trials, best score, and optimization configuration — same format as sklearn optimized models.

v2.0 CV Report Format: The exported CV table shows Entrene Score and Pruebe Score in separate columns, with Mean and Std on separate rows. For LOO-CV, the report shows R^2 (global) at the top and a per-sample MAE table below. Ejemplo:

```
Cross-Validation (5-Fold):
    Fold |  Entrene Score |   Pruebe Score
  -------+--------------+-------------
       1 |     0.999960 |     0.975628
       2 |     0.999960 |     0.973907
     ...
    Mean |     0.999962 |     0.949832
     Std |     0.000003 |     0.050600
```

Pestana de Ensamblado

The Pestana de Ensamblado allows you to combine multiple models for improved predictions.

Seleccionar Modelos

103. Haga clic "Actualice Lista de Modelos" to load available models

104. Seleccione multiple models from the list (Ctrl+Haga clic for multiple)
105. Available sources:

- Entreneed models from Pestana de Entrenamiento
- Optimiced models from Pestana de Optimizacion

Configuracion de Ensamblado

- Ensemble Escriba:
- Averaging: Mean of all model predictions
- Voting: Weighted voting (for classification)
- Weighted: Weighted average (equal weights by default)
- Number of Models: Shows how many models are selected

Validacion Cruzada (v2.0)

Ensembles now support the same five CV strategies as sklearn regressors (see Elegir una Estrategia de VC): K-Fold, Stratified K-Fold (quantiles, with optional reserved bin), Agrupe K-Fold, LOO, and LOGO. The chosen strategy is used both when training the ensemble and when optimizing the ensemble weights (weight optimization shares the strategy/groups but uses its own fold count). Agrupeed strategies keep groups intact, so the ensemble's reported score is free of chemical/batch leakage. Ensembles are most reliable with N ≥ 100.

Crear Ensamblado

106. Seleccione 2 or more models
107. Elija ensemble type
108. Haga clic "Cree Ensemble"
109. Wait for evaluation to complete

Resultados de Ensamblado

Despues creation, you'll see:

- Ensemble Name: Auto-generated identifier
- Pruebe Metrics: RMSE, MAE, R^2 on test set
- Entreneing Metrics: RMSE, MAE, R^2 on training set

Guardar Ensamblados

Haga clic "Guarde Ensemble Model":

110. Elija format (Bundle or Legacy)
111. Seleccione destination folder
112. Ensemble appears in Pestana de Resultados and can be used in Explainability

Pestana de Explicabilidad

The Pestana de Explicabilidad provides model interpretation tools including SHAP values, permutation importance, and partial dependence plots.

Seleccion de Modelo

113. Haga clic "Actualice Lista de Modelos" to load available models
114. Seleccione a model from the dropdown
115. Available models: Entreneed, Optimiced, Neural Network, Ensemble

v2.0 FIX: Redes Neuronales no longer appear duplicated. Each NN appears once as "Neural Network (Neural Network)".

Opciones de Analisis

Seleccione which analyses to run:

- Compute Valores SHAP: Feature contributions using SHAP
- Compute Permutation Importeance: Feature importance via permutation
- Compute Dependencia Parcial Plots: Feature effect visualization
- Analice Aplicabilidad de Dominio: Data coverage analysis

Numero de Caracteristicas

Establezca how many top features to display (1-100, default: 10)

Ejecutar Analisis

116. Seleccione model and options
117. Haga clic "Ejecute Explainability Analysis"
118. Monitoree progress in the log area
119. Resultados appear in tabs below

v2.0 NOTE: The log area now shows detailed diagnostics for each importance type (sum, all_zero, all_nan) to help verify results.

Pestanas de Resultados

Importeancia de Caracteristicas

Tabla showing:

- Feature: Feature name
- Model Importeance: Built-in importance (if available) or permutation-based surrogate
- Permutation: Permutation importance score
- SHAP: Mean absolute SHAP value
- Average Rank: Combined ranking across methods

Notas:

- For Redes Neuronales, Model Importeance shows "N/A" as they don't have built-in feature importance
- For HistGradientBoosting and LightGBM: Model Importeance is computed via a fast permutation-based surrogate (v2.0). Anteriorly these showed zeros due to oversubscription.
- For SVR and Ridge: Model Importeance comes from coefficients or permutation; random_state/n_jobs are hidden from ManualParamCuadro de dialogo as these models are deterministic

Valores SHAP

Summary of SHAP analysis:

- Global feature importance (top N features)
- Mean absolute SHAP value per feature

v2.0 SHAP Explainer Seleccioneion (automatic, no user action needed):

- TreeExplainer for tree models (Bosque Aleatorio, XGBoost, LightGBM, Histogram Gradient Boosting, Gradient Boosting, CatBoost) — exact and fast
- LinearExplainer for linear models (Ridge, Logistic Regression) — exact
- DeepExplainer for Redes Neuronales — specialized for PyTorch
- KernelExplainer only as fallback for generic models (SVR, k-NN, Proceso Gaussiano) — slower, approximate

v2.0 Performance: Explainability uses smart parallelization — single-threaded models (SVR, Ridge) use all CPU cores for permutation importance, while multi-threaded models (RF, XGB, LGBM, HistGB) use one core to avoid oversubscription. On PyInstaleer Ventanas builds, parallelization is always disabled.

Dependencia Parcial

Lista of features analyzed with partial dependence plots.

Nota: Dependencia Parcial Plots are not available for Redes Neuronales due to sklearn limitations.

Aplicabilidad de Dominio

Analysis of model applicability domain:

- Entreneing Samples: Number of training samples
- Original Features: Number of input features
- PCA Components: Number of PCA components used
- Explained Variance: Variance captured by PCA
- Thresholds: Leverage, Mahalanobis, k-NN distance thresholds

Explicacion de Prediccion

Explain individual predictions:

120. Ingrese instance index (row number)
121. Haga clic "Explain Predigaion"
122. Vea:

- Predigaed value
- SHAP contributions (top features)
- Top positive/negative contributors

Exportear Resultados

Despues running an explainability analysis, click "Exporte Resultados as Text" to save all results to a file:

123. Haga clic the "Exporte Resultados as Text" button (below "Ejecute Explainability Analysis")
124. Elija a save location (default filename: explainability_{model_name}_{timestamp}.txt)
125. The exported file includes:

- Header: Model name, timestamp, and report title
- Section 1 — Importeancia de Caracteristicas: Complete table with Feature, Model Importeance, Permutation, SHAP, and Average Rank
- Section 2 — Valores SHAP: Global feature importance summary
- Section 3 — Dependencia Parcial: Lista of analyzed features (or NN unavailable message)
- Section 4 — Aplicabilidad de Dominio: Entreneing samples, features, PCA components, explained variance, thresholds
- Section 5 — Explicacion de Prediccion: Unico-instance explanation (if generated)
- Section 6 — Analysis Registre: Complete diagnostic log from the analysis run

Consejo: Exporte after each analysis to keep a permanent record of model interpretation results.

Pestana de Prediccion

The Pestana de Prediccion allows you to make predictions on new data using saved models.

Cargar un Modelo

126. Haga clic "Cargue Model Bundle"
127. Seleccione the folder containing:

- manifest.json (required)
- pipeline.joblib or model.pt (required)
- preprocessor.joblib (if applicable)

The model information will be displayed, including:

- Model type and algorithm
- Feature names expected
- Target column name
- Model metrics (if available)
- Valided with (v2.0): the cross-validation strategy used during training, read from the bundle manifest (e.g. "Agrupe K-Fold (by chemical system)"). This is a read-only traceability line — prediction never re-runs CV.

Nota on scope: Cross-validation (stratified, grouped, LOO, LOGO) is a training-time evaluation technique. It does not apply to prediction on new data, so the Prediga tab has no CV controls — it only reports how the loaded model was validated.

v2.0: Modelos con Estructuras Vinculadas

Cuando you load a model that was trained with linked crystal or molecule features, the Pestana de Prediccion automatically shows the "Vincule Structures for Predigaion (v2.0)" section.

Como Funciona

128. Auto-detection: The system reads the model's manifest and detects if it requires linked structures (crystal_ or mol_ features)
129. Structure type display: Shows "Crystal (CIF)" or "Molecule (.xyz)" based on the model
130. Validation message: Cuando linked features are missing, the status shows "Linked features required: Use 'Prediga with Linked Structures' section below"

Pasos para Prediccion Vinculada

131. Cargue the model bundle (trained with linked features)
132. Cargue your CSV data (must include an ID column matching structure filenames)
133. In the "Vincule Structures" section:

- Haga clic "Navegue..." to select the folder with .cif or .xyz files
- Seleccione the ID Column from the dropdown (must match structure filenames)
- Verifique the Escriba is correct (auto-detected from model)

134. Haga clic "Prediga with Linked Structures"
135. The system will:

- Match CSV IDs with structure files
- Extract structural features
- Auto-generate composition features if the model was trained with them
- Ejecute prediction
- Report: "Exitofully linked structures and predicted N rows"

Los 2 Casos Principales

Case	Structure Escriba	Que You Need	Que Happens
A	Crystal (CIF)	CSV with IDs + folder with .cif files	Vincule .cif by ID → 45 features: 11 structural + 34 periodic compositional (electronegativity, ionization, oxidation states, etc.) → predict
B	Molecule (.xyz)	CSV with IDs + folder with .xyz files	Vincule .xyz by ID → 149 features: composition + geometry 3D + topology + 34 periodic properties + oxidation states → predict

Crystal vs. Molecule — which do I have?

- Crystal = periodic structure with a unit cell (metals, oxides, minerals, semiconductors). File format: .cif primarily
- Molecule = isolated structure without a unit cell (organic molecules, ligands, solvents, gases). File format: .xyz
- Ambos linkers extract 34 periodic properties (electronegativity, ionization energy, atomic radius, oxidation states, etc.) automatically from the structure file

Cargar Archivos de Estructura Directamente (Sin CSV)

You can also train directly on structure files without a CSV:

136. Pestana de Datos → Haga clic "Cargue Molecular Structures" or "Cargue Molecular Directory" (for .xyz)
137. Pestana de Datos → Haga clic "Cargue Crystal Structures" or "Cargue Crystal Directory" (for .cif)
138. Features are extracted automatically — select target and train

Que Ocurre Sin Vinculacion

Si you click "Ejecute Predigaion" (the regular button) with a linked model but without providing structure files:

- An informative dialog appears explaining what's needed
- Shows: "This model was trained with linked [crystal/molecule] features that are missing from the input data"
- Provides the exact steps to use the "Prediga with Linked Structures" section
- No crash — graceful error with clear next steps

Cargar Datos de Prediccion (Modelos Simples)

139. Haga clic "Cargue Predigaion Data"
140. Seleccione your CSV or Excel file
141. Data preview will appear

Validacion de Caracteristicas

The system checks if all required features are present:

- Green: Todos features present
- Blue: Linked features required (use "Prediga with Linked Structures")
- Yellow: Missing features can be auto-generated
- Red: Missing features cannot be generated

Auto-Generar Caracteristicas

Si your data has a formula column (e.g., "composition", "formula"):

142. Marque "Auto-generate composition features from formula column if missing"
143. The system will generate composition descriptors automatically

Realizar Predicciones (Modelos Simples)

144. Ensure all required features are present (green status)
145. Haga clic "Ejecute Predigaion"
146. Predigaions appear in the table

Intervalos de Confianza

Marque "Include Intervalos de Confianza" to get prediction intervals:

- Regression: Based on CV residual analysis (68% and 95% quantiles)
- Stored as lower_bound and upper_bound columns

Exportear Predicciones

147. Haga clic "Exporte Predigaions"

148. Elija save location
149. Predigaions are saved as CSV with:

- Todos original columns
- Nuevo prediction column
- lower_bound and upper_bound (if confidence intervals enabled)

Pipeline de Fuga de Datos Cero (v2.0)

This section explains the internal architecture change that makes AutoAIM Studio's cross-validation scientifically valid.

Cual era el Problema?

In most AutoML tools (including AutoAIM before v2.0), the typical workflow looks like this:

```
1. Cargue data
2. Fit StandardScaler on ALL training data
3. Transform training data
4. Ejecute cross-validation on the already-scaled data
```

The bug: Paso 2 sees the validation data. Cuando the scaler computes the mean and standard deviation, it includes the validation fold. Durante CV, that fold is supposed to be "unseen," but its statistics have already leaked into the scaler.

This produces optimistically biased CV scores — the model appears better than it really is because the validation data indirectly influenced the preprocessing.

Como AutoAIM v2.0 lo Corrige

AutoAIM now uses sklearn Pipeline internally:

```
1. Cargue data → keep it RAW (no scaling)
2. Compile Pipeline([preprocessor, model])
3. Ejecute cross_validate(pipeline, X_raw, y, cv=...)
```

Inside cross_validate(), sklearn does this for every fold:

```
For fold i:
  preprocessor.fit(X_train_fold)   ← scaler sees ONLY training data
  X_train_scaled = preprocessor.transform(X_train_fold)
  model.fit(X_train_scaled, y_train_fold)
  X_val_scaled = preprocessor.transform(X_val_fold)
  score = evaluate(model, X_val_scaled, y_val_fold)
```

Result: The validation fold is truly unseen during both preprocessing and model training.

Donde se Aplica?

Component	Pipeline Used?	Details
Pestana de Entrenamiento	Yes	Todos sklearn models trained with preprocessor parameter
Optimization (Bayesian)	Yes	Each trial wraps estimator in Pipeline before CV
Optimization (Cuadricula Busque)	Yes	GridBusqueCV runs on Pipeline; param_grid prefixed with model__
Ensemble	Yes	Base models are already Pipelines from Entreneing
Neural Network	Manual	CV loop clones and refits preprocessor per fold before training each PyTorch network
Saved Models	Yes	Exporteed bundle contains the fitted Pipeline (preprocessor + model)
Pestana de Prediccion	Yes	Inference receives raw DataFrame; Pipeline applies preprocessing internally

Que Significa para los Usuarios

You don't need to do anything differently. The GUI looks and works exactly the same. The change is entirely internal:

- Cargue your data normally
- Seleccione target, preprocess, generate features — same as before
- Entrene models — CV scores are now valid (no hidden leakage)
- Guarde models — the bundle contains a complete Pipeline
- Prediga — pass raw data; the Pipeline handles scaling automatically

Para Desarrolladores

Si you're extending AutoAIM programmatically:

```
from app.core.data_manager import DataManager
from app.core.trainer import ModelEntreneer
```

```
from sklearn.ensemble import RandomForestRegressor

dm = DataManager()
dm.load_data("data.csv")
dm.set_target_column("target")

# Returns RAW DataFrames (v2.0)
X_train, X_test, y_train, y_test = dm.prepare_data()

# Get the unfitted preprocessor
preprocessor = dm.get_preprocessor()

# Entrene with Pipeline (zero data leakage)
trainer = ModelEntreneer()
result = trainer.train(
    RandomForestRegressor(), "RF",
    X_train, y_train, X_test, y_test,
    problem_type="regression",
    preprocessor=preprocessor,  # ← key: enables Pipeline
)

# result.model is a fitted Pipeline
# Accepts raw DataFrames for prediction
preds = result.model.predict(X_test)
```

Si you omit preprocessor, the legacy path (numpy arrays, no Pipeline) still works for backward compatibility.

Impacto Cientifico

This fix ensures that:

150. CV scores reflect true generalization — not an optimistic upper bound
151. Model comparisons are fair — all models evaluated under identical no-leakage conditions
152. Published results are defensible — the methodology follows best practices in ML for materials science
153. Saved models are deployment-ready — the Pipeline encapsulates the complete preprocessing logic

Vinculacion de Estructuras

Structure linking bridges tabular datasets with molecular or crystal structure files, enriching your data with structural descriptors before training. AutoAIM Studio supports two types of structure linking:

Feature	Crystal Structures	Molecular Structures
File format	.cif	.xyz
ID matching	Filename = ID value	Filename = ID value
Features extracted	149 crystal descriptors	149 molecular descriptors
Dependencies	pymatgen	numpy only
Module	CrystalFeatureExtractor	MoleculeFeatureLinker

Vinculacion de Estructuras Cristalinas (CIF)

Como Funciona

154. Your dataset has an ID column (e.g., material_id) matching CIF filenames
155. CIF files are named {material_id}.cif in a folder
156. The linker extracts 149 crystal descriptors: lattice (~10), composition (~13), structural (~3), Magpie (106), advanced (~15), derived (~2). Same features as direct CIF loading, with crystal_ prefix.
157. Features are merged into your dataset as crystal_* columns

Verificacion Pre-Vinculacion

Antes linking, the system reports:

- Matched: Como many IDs have corresponding CIF files
- Missing: Como many IDs lack CIF files (features filled with NaN)
- Duplicates: Warns if multiple rows share the same ID

Importeantee: Deteccion de No Cristalinos

The system detects isolated molecules in artificial simulation boxes and warns you that crystal descriptors will be physically meaningless for these structures.

Vinculacion de Estructuras Moleculares (.xyz)

Como Funciona

158. Your dataset has an ID column matching .xyz filenames
159. .xyz files are named {id}.xyz in a folder

160. The linker extracts three levels of molecular descriptors:

Level 1 — Composition (~20 features)

- Atom counts per element (H, C, N, O, S, P, F, Cl, Br, I, Si, B)
- Molecular mass, heavy atom count, elemental diversity
- Element ratios: H/C, O/N, C/N, S/C, P/C, halogen ratios

Level 2 — Geometry 3D (~20 features)

- Centre of mass, radius of gyration
- Bounding box: volume, surface, diagonal
- Principal moments of inertia + oblate/prolate shape index
- Asphericity, spherical defect
- Pairwise distance statistics (mean, std, min, max, median)

Level 3 — Topology (~45 features)

- Bond estimation, coordination numbers
- Bond angle statistics + 10-bin histogram
- Ring detection (3-8 membered) + ring size histogram
- Functional group detection: OH, NH, C=O, C=C, C≡C, C≡N, C-F, C-Cl, C-Br, PO, CS, SO
- Bond type ratios, saturation/unsaturation indices

Ventajas

- Zero external dependencies: Pure numpy, no RDKit
- Numeric prefix tolerant: Handles .xyz files with prefixed elements (e.g., "1C 0.0 0.0 0.0")
- Missing structures filled with NaN (trainable after imputation)

Predecir con Modelos de Estructuras Vinculadas

El Problema

Si you train a model with linked crystal/molecule features and later try to predict on data without those features, the system detects this and provides an actionable error instead of crashing.

La Solucion

Cuando a model requiring linked structures is loaded:

161. The Pestana de Prediccion shows a "Vincule Structures" section
162. Seleccione your structure folder and ID column
163. Haga clic "Prediga with Linked Structures"

The system automatically:

164. Matches IDs with structure files

165. Extracts structural features
166. Combines them into your data
167. Runs prediction

Uso Programatico

```
from app.core.inference_engine import InferenceEngine

engine = InferenceEngine("path/to/model_bundle")

# Marque if model requires linked structures
if engine.is_structure_linked_model():
    reqs = engine.get_linked_structure_requirements()
    print(f"Requires: {reqs['structure_type']} features")
    print(f"Missing: {reqs['missing_feature_count']} features")

# Prediga with automatic linking
result = engine.predict_with_linked_structures(
    df=new_data,
    id_column="material_id",
    structure_folder="/path/to/cif_or_xyz_files/"
    # structure_type auto-detected from manifest
)
```

Que Ocurre Sin Vinculacion

Si you call predict() without linked features:

- Informative error: "This model requires linked crystal features... Use predict_with_linked_structures()"
- Ejemplo code: Shows exact syntax needed
- No crash: Graceful error with clear next steps

Solucion de Problemas

Problemas Comunes

"Cannot generate missing features automatically"

Cause: Missing composition features and no formula column detected

Solution:

168. Ensure your prediction data has a column with chemical formulas
169. Marque the "Auto-generate composition features" checkbox

170. The column name should contain "formula", "composition", or "structure"

"Seleccioneed model does not have predict method"

Cause: Model object is corrupted or not properly loaded

Solution:

171. Retrain the model
172. Marque if model was saved correctly
173. For Redes Neuronales, ensure model was built before training

"Model 'X' not found in any source"

Cause: Model was deleted or not properly stored

Solution:

174. Retrain the model
175. Guarde the model again after training

"SHAP values not available for this model"

Cause: SHAP computation failed (usually for Redes Neuronales)

Solution:

176. Ensure model has a working predict method
177. Try with a smaller dataset sample
178. Marque console for detailed error messages

"Partial dependence plots not available"

Cause: PDP is not supported for Redes Neuronales

Solution:

- This is expected behavior. Use Importeancia de Caracteristicas and SHAP tabs instead.

Problemas de Estrategia de Validacion Cruzada (v2.0)

"Agrupeing produced N group(s) but K folds were requested"

Cause: With Agrupe K-Fold, you asked for more folds than there are groups.

Solution: Reduce the CV Folds to a number ≤ the number of groups, or choose a different grouping (e.g. "Exact formula" or a higher cluster count produces more groups than "Chemical system").

"Leave-One-Agrupe-Out needs at least 3 groups"

Cause: LOGO requires at least 3 distinct groups.

Solution: Use Agrupe K-Fold instead, or pick a grouping that yields more groups.

"N sample(s) have no value in column 'X'"

Cause: You chose "Dataset column" grouping, but that column has missing values.

Solution: Fill or remove those rows first. AutoAIM never silently drops samples.

"No composition/formula column was detected"

Cause: You chose a chemical grouping source (chemical system, formula, or clustering), but no column in your dataset looks like chemical formulas.

Solution: Agregue a formula/composition column, or use "Dataset column" grouping.

"Could not build quantile bins..."

Cause: Stratified K-Fold couldn't bin the target so every fold sees every bin (target nearly constant, or too few samples for the requested folds).

Solution: Reduce CV Folds, or use standard K-Fold. Si the target is constant, stratification is impossible.

"Cannot reserve a bin for X: only N sample(s) match..."

Cause: You reserved a bin for a special value, but fewer samples match it than the number of folds, so the value cannot appear in every fold.

Solution: Increase the Tolerance (if your data stores the value as e.g. 0.01 instead of 0), reduce CV Folds, or turn off the reserved bin and use plain quantile stratification.

Reserved bin warning: "matches only N of M samples (X%)"

Cause: Your special value is a small fraction of the data (<5%), so the reserved bin may have little effect — or it may not be a true 'property absent' marker.

Solution: Confirme the value is a genuine degenerate marker (e.g. Tc = 0 K for non-superconductors, Eg = 0 eV for metals). Si it is, proceed; if not, disable the reserved bin.

Consejos de Rendimiento

Acelerar el Entrenamiento

179. Reduce CV folds: Use 3 instead of 5
180. Use GPU: Habilite GPU acceleration for Redes Neuronales
181. Reduce trials: Use fewer optimization trials (20-30 instead of 50+)
182. Sample data: Entrene on a subset for initial exploration

Mejorar la Precision

183. Generate composition features: Use all 106 descriptors
184. Optimice hyperparameters: Ejecute optimization with more trials
185. Cree ensembles: Combine multiple models
186. Try different algorithms: Not all algorithms work equally well for all datasets

Reducir el Uso de Memoria

187. Deshabilite composition features: Si not needed
188. Reduce batch size: For Redes Neuronales
189. Use fewer optimization trials: Each trial keeps results in memory

Obtener Ayuda

Si you encounter issues not covered here:

190. Marque the log output in each tab
191. Review the console/terminal for detailed error messages
192. Ensure your data is properly formatted (CSV/Excel)
193. Verifique all required columns are present

Referencia de Caracteristicas

Caracteristicas de Composicion (106 Descriptores)

The custom composition feature generator computes the following statistics for each chemical formula:

Atomic Properties:

- Atomic number (mean, std, min, max, range, median, percentiles)
- Atomic mass (mean, std, min, max, range, median, percentiles)
- Covalent radius (mean, std, min, max, range, median, percentiles)
- Van der Waals radius (mean, std, min, max, range, median, percentiles)

Electronic Properties:

- Electronegativity (mean, std, min, max, range, median, percentiles)

- Ionization energy (mean, std, min, max, range, median, percentiles)
- Electron affinity (mean, std, min, max, range, median, percentiles)

Physical Properties:

- Density (mean, std, min, max, range, median, percentiles)
- Boiling point (mean, std, min, max, range, median, percentiles)
- Melting point (mean, std, min, max, range, median, percentiles)

Periodic Tabla Properties:

- Agrupe number (mean, std, min, max, range, median, percentiles)
- Period number (mean, std, min, max, range, median, percentiles)

Additional Features:

- Number of unique elements
- Total number of atoms
- Element fractions

These features are computed from elemental properties and provide a comprehensive representation of material composition for machine learning models.

Apendice: Algoritmos de Modelos

Bosque Aleatorio

- Escriba: Ensemble of decision trees
- Strengths: Handles non-linear relationships, robust to outliers
- Best for: General-purpose regression, feature importance

Gradient Boosting

- Escriba: Sequential ensemble
- Strengths: Alto accuracy, handles complex patterns
- Best for: Competitive accuracy, medium-sized datasets

XGBoost

- Escriba: Optimiced gradient boosting
- Strengths: Fast, regularized, handles missing values
- Best for: Tabular data competitions, production systems

LightGBM

- Escriba: Histogram-based gradient boosting
- Strengths: Muy fast, memory efficient
- Best for: Large datasets, speed-critical applications

CatBoost

- Escriba: Gradient boosting with ordered boosting
- Strengths: Excellent categorical handling, reduced overfitting
- Best for: Datasets with many categorical features

SVR (Regresion por Vectores de Soporte)

- Escriba: Kernel-based method
- Strengths: Effective in high dimensions, memory efficient
- Best for: Small to medium datasets, non-linear relationships

Regresion Ridge

- Escriba: Linear regression with L2 regularization
- Strengths: Simple, fast, prevents overfitting
- Best for: Baseline model, linear relationships

Redes Neuronales

- Escriba: Deep learning with customizable architecture
- Strengths: Learns complex non-linear patterns
- Best for: Large datasets, complex feature interactions

Documentacion de Caracteristicas

For the complete list of every descriptor computed by AutoAIM, see the dedicated feature reference documents:

Document	Features	Description
FEATURES_CRYSTAL.md	149	Crystal structure descriptors (lattice, composition, Magpie, structural, advanced)
FEATURES_MOLECULAR.md	149	Molecular descriptors (composition, geometry 3D, topology, oxidation states)
FEATURES_COMPOSITION.md	106	Magpie composition statistics (13 properties × 8 stats + 2 derived)

Citacion

Si usa AutoAIM Studio en su investigacion, por favor cite:

```
@article{vera2026autoaim,
  title = {AutoAIM Studio: A desktop application for automated
machine learning in materials science},
  author = {Vera De la Garza, G. V. and Fomine, S.},
  journal = {SoftwareX},
  volume = {35},
  pages = {102945},
  year = {2026},
  doi = {10.1016/j.softx.2026.102945},
  url = {https://github.com/cesargabrielvera1-ux/AutoAIM-Studio}
}
```

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